

ARM PrimeCell

Multiport Memory Controller (PL172)

Errata Notice

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This document contains all errata known at the date of issue in releases from revision r2p2 up to and including revision.r2p3. Errata relating to previous releases are in a separate document.

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- The document title
- The documents number
- The page number(s) to which your comments refer
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General suggestion for additions and improvements are also welcome.

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INTRODUCTION

Scope

This document describes errata categorised by level of severity. Each description includes:

- where the implementation deviates from the specification
- the conditions under which erroneous behaviour occurs
- the implications of the erratum with respect to typical applications
- the application and limitations of a 'work-around' where possible

Categorisation of Errata

Errata recorded in this document are split into three levels of severity:

- Category 1** Features which are impossible to work around and severely restricts the use of the product in all or the majority of applications rendering the device unusable.
- Category 2** Features which contravene the specified behaviour and may limit or severely impair the intended use of specified features but does not render the product unusable in all or the majority of applications.
- Category 3** Features that were not the originally intended behaviour but should not cause any problems in applications.

Change Control

Changes in Document v11

Page	Status	ID	Cat	Summary
p17	Updated	309459	Doc	UserGuide States Use Denali*.mk Instead of DirSimMakefile*.mk
p15	Updated	316115	Doc	Update to Errata 3.9
p12	Updated	323940	Cat 3	Burst mode timing
p11	Updated	284509	Cat 2	CKE signal is low at power-up.
p15	New	324261	Doc	Incorrectly logged Errata 3.7

Changes in Document v7

Status	ID	Cat	Summary
Updated	309459	Doc	UserGuide States Use Denali*.mk Instead of DirSimMakefile*.mk
Updated	314789	Doc	Synthesis should be HCLK:MPMCCLK = 1:1
Updated	280959	Cat 3	Incorrectly shipped Tests and Log Files

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Updated	284509	Cat 2	CKE signal is low at power-up.
Updated	319664	Cat 1	Some SDRAM writes missed and not written to memory

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ERRATA SUMMARY TABLE

The errata associated with this product affect product versions as follows:

<i>ID</i>	<i>Cat</i>	<i>Summary</i>	<i>r2p1- 00LTD0</i>	<i>r2p2- 00LTD0</i>	<i>r2p2- 00LTD1</i>	<i>r2p3- 00REL0</i>
280959	Cat 3	Incorrectly shipped Tests and Log Files		X		
284509	Cat 2	CKE signal is low at power-up.		X	X	
309459	Doc	UserGuide States Use Denali*.mk Instead of DirSimMakefile*.mk		X	X	
314789	Doc	Synthesis should be HCLK:MPMCCLK = 1:1		X	X	
316115	Doc	Update to Errata 3.9	X			
319664	Cat 1	Some SDRAM writes missed and not written to memory		X	X	
323940	Cat 3	Burst mode timing	X			
324261	Doc	Incorrectly logged Errata 3.7	X	X	X	

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ERRATA - CATEGORY 1

[319664](#): Some SDRAM writes missed and not written to memory

Status

Affects: product PL172_MPMC.
Fault raised on: revision r2p2 00LTD0 of PL172_MPMC
Fault status: Present in: r2p2-00LTD0,r2p2-00LTD1, Fixed in r2p3-00RELO

Description

This issue can occur when any burst that does not fill the buffer completely takes place, that is when either:

- all buffers are occupied, with one of them containing partial data

or

- three buffers are full(with writes) and a fourth buffer is allocated data from a partial write (due to a buffer miss)

The scenario that can produce the failure is as follows

1) Since each buffer is QQword deep, if a write that does not fill a QQWord occurs, it will result in a partially filled buffer. This write can be any transaction that writes one or two data elements into the buffer.

Once the write is complete from AHB, the data will remain in the buffer, and is flushed to the memory at the earliest opportunity.

2) In the meantime, if a Read is issued to the location where the partial write occurred, i.e. a buffer hit, the read can be serviced from the buffer. But if the read is a burst read (for example an INCR8), all the data requested by the read may not be present in buffer and hence data needs to be fetched from memory.

3) As soon as the data for the remaining AHB burst arrives at the buffer, it is made available to the AHB, until the AHB burst read is complete (for an INCR8 1 from the buffer and 7 from external memory).

4) However, a read posted by the buffer may take longer to complete.

Thus as far as the AHB is concerned it has the data it needs for the read and can proceed to the next transaction, but the buffer would be busy fetching the remaining data from memory.

NOTE:

In the example of an INCR8, it might require 12 fetches from external memory, only if the address posted to the Dynamic memory controller from the buffer is an unaligned address in memory.

5) If the new AHB transaction then tries to carry out a write to the buffer that is currently being serviced (from memory), then HREADY will be taken LOW for this write, holding off this AHB write, as no two agents (memory controller and AHB) can access the same buffer at the same time.

6) Whilst the memory controller (dynamic state machine) is finishing off fetching data from memory from the remaining burst to the buffer, it changes its status to NOT-BUSY and one of the data state machines (as there are two) moves to IDLE and can accept a new command.

7) In accepting a new command the buffer block flushes one of the other buffers that were full and a write to external memory is scheduled.

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This buffer flush thus results in wrong buffer status information inside the MPMC which returns an HREADY HIGH on the AHB which was pending with a write to the buffer that is currently being serviced (read) from memory.

8) The MPMC accepts this new write request but it never gets written into either the buffer or memory - effectively the data is lost.

However, this loss of data will only be detected when an attempt is made to read the data written in Step 8, which will return the data that was in the memory prior to Step 8.

Implications

Some SDRAM writes can be missed

Workaround

The modification consists of an additional an AND gate. Modification:

File: MpmcBufCntrl.vhd

1) Addition to sensitivity list in p_BufToAHBComb process. (Addition of FlushInWrQ)

Original code:

```
p_BufToAHBComb : process (ValidArbQ, iUnBufMemAcc, BufSelected, AHBWrReq,  
    ByteMatchC10, ByteMatchC11, ByteMatchC12,  
    ByteMatchC13, BufRdData0, BufRdData1, BufRdData2,  
    BufRdData3, AHBRdReq, HitToNotLkdBuf, OneBufEmpty,  
    FlushCycle, BufHit0, BufHit1, BufHit2, BufHit3,  
    BufCanBeUsed, BufEmpty0Q, BufEmpty1Q, BufEmpty2Q,  
    BufEmpty3Q)
```

Updated code:

```
p_BufToAHBComb : process (ValidArbQ, iUnBufMemAcc, BufSelected, AHBWrReq,  
    ByteMatchC10, ByteMatchC11, ByteMatchC12,  
    ByteMatchC13, BufRdData0, BufRdData1, BufRdData2,  
    BufRdData3, AHBRdReq, HitToNotLkdBuf, OneBufEmpty,  
    FlushCycle, BufHit0, BufHit1, BufHit2, BufHit3,  
    BufCanBeUsed, BufEmpty0Q, BufEmpty1Q, BufEmpty2Q,  
    BufEmpty3Q, FlushInWrQ) 2) Addition of another term to the if statement in the  
p_BufToAHBComb process. (Addition of AND (FlushInWrQ = '0'))
```

Original code:

```
if ((ValidArbQ = '1') and (iUnBufMemAcc = '0') and  
    ((AHBWrReq = '1') and (((HitToNotLkdBuf = '1') or  
    (BufCanBeUsed = '1')) or
```

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((OneBufEmpty = '0') and (FlushCycle = '1')))) then

Updated code:

if ((ValidArbQ = '1') and (iUnBufMemAcc = '0') and
((AHBWrReq = '1') and (((HitToNotLkdBuf = '1') or
(BufCanBeUsed = '1')) or
((OneBufEmpty = '0') and (FlushCycle = '1') AND (FlushInWrQ = '0')))) then

=====

||

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ERRATA - CATEGORY 2

[284509](#): CKE signal is low at power-up.

Status

Affects: product PL172_MPMC.
Fault raised on: revision r2p2 00LTD0 of PL172_MPMC
Fault status: Present in: r2p2-00LTD0,r2p2-00LTD1, Fixed in r2p3-00RELO

Description

At power-up the CKE signals are low. Some memory devices require both the DQM and CKE signals to be high otherwise they may drive the DQ signals and cause bus contention if multiple devices are present.

Implications

Bus contention could occur at power-up, until the initialisation process of the SDRAM has completed.

Workaround

- 1) Use SDRAM devices that do not require the DQMs to be high at power-up or use only a single bank of devices.
- 2) The following RTL modification can be made. Please note that this change has not been validated at ARM, and so an extra test to test this change will have to be created. The existing test vectors will produce errors when simulating as they will be expecting the CKE to be LOW on reset.

- in MpmcDyIntRegBlk.v/vhd (line 2442): iDyCKEOut must be set (instead of cleared) when nPOR = '0' :

```
p_CKESeq : process (MPMCCLK, nPOR)
begin
    if (nPOR = '0') then
--      iDyCKEOut      <= "0000";   old line
      iDyCKEOut      <= "1111"; -- new line
    elsif (MPMCCLK'event and MPMCCLK = '1') then
      iDyCKEOut      <= NxtDyCKEOut;
    end if;
end process p_CKESeq;
```

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ERRATA - CATEGORY 3

[323940](#): Burst mode timing

Status

Affects: product PL172_MPMC.
Fault raised on: revision r2p1 00LTD0 of PL172_MPMC
Fault status: Present in: r2p1-00LTD0, Fixed in r2p2-00LTD0

Description

When the burst mode static memories are accessed with the value of the page mode read access time programmed as 0, the 5th read in the burst is also issued with a wait-page timing and the wait-read timing is used for the 6th read instead of the 5th

Implications

none

Workaround

The fix for this issue is in the file MpmcStExtIf.v as listed below:

Change the following lines

```
--
-----
-- BmlenEnd generation logic to determine the end of the Burst length
-- and the
-- crossover of the address Quad-boundary for page mode enabled devices.
-- This logic is used for the purpose of introducing the burst length
-- restriction. Only burst length of four is supported for page mode
-- enabled
-- devices
--
-----
p_BmlenComb : process (iPgLdEnd, StMemWidthQ, iStAddr, RdCntLd,
                      StPgModeRdQ, MpmcStStateMRd)
begin
  NxtPgLdEnd      <= iPgLdEnd;
  if (StPgModeRdQ = '1') then
    if (RdCntLd = '1') then
      NxtPgLdEnd      <= '0';
    elsif (((StMemWidthQ = "00" and iStAddr(1 downto 0) = "11") or
            (StMemWidthQ(0) = '1' and iStAddr(2 downto 1) = "11") or
            (StMemWidthQ(1) = '1' and iStAddr(3 downto 2) = "11")) and
            (MpmcStStateMRd = '1')) then
      NxtPgLdEnd      <= '1';
    end if;
  end if;
end process p_BmlenComb;
To
```

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```
-----
-- BmlenEnd generation logic to determine the end of the Burst length
and the
-- crossover of the address Quad-boundary for page mode enabled devices.
-- This logic is used for the purpose of introducing the burst length
-- restriction. Only burst length of four is supported for page mode
enabled
-- devices
--
-----
p_BmlenComb : process (iPgLdEnd, StMemWidthQ, NxtStAddr, RdCntLd,
                      StPgModeRdQ, MpmcStStateMRd)
begin
  NxtPgLenEnd      <= iPgLdEnd;
  if (StPgModeRdQ = '1') then
    if (RdCntLd = '1') then
      NxtPgLenEnd    <= '0';
    elsif (((StMemWidthQ = "00" and NxtStAddr(1 downto 0) = "11") or
            (StMemWidthQ(0) = '1' and NxtStAddr(2 downto 1) = "11") or
            (StMemWidthQ(1) = '1' and NxtStAddr(3 downto 2) = "11")) and
            (MpmcStStateMRd = '1')) then
      NxtPgLenEnd    <= '1';
    end if;
  end if;
end process p_BmlenComb;
-----
```

Please note: that due to the differences in the RTL on which these fixes have been applied locally, and the RTL on which you will be applying the fixes, a possibility of unexpected behaviour cannot be ruled out. This information is provided with the understanding that these fixes are done as is. These may need to be applied with proper knowledge of the implications to your system.

[280959](#): Incorrectly shipped Tests and Log Files

Status

Affects: product PL172_MPMC.
Fault raised on: revision r2p2 00LTD0 of PL172_MPMC
Fault status: Present in: r2p2-00LTD0, Fixed in r2p2-00LTD1

Description

- 1) There is a BankCross Test that exists in the verification/bustest directory that is not documented. This test is an obsoleted test that should not have been shipped. This test should be ignored.
- 2) The MpmcTrAhbif.vhd file does not compile using Modelsim version 5.6e. There is an improper concatenation within that file.
- 3) The MultiPortDiffComb8.c file does not compile as it has errors. More specifically, the variables LoBits and Bound are not declared properly.
- 4) The Mpmc.c file has a typo that causes the LatencyTest10.c to not be compiled.
- 5) The LatencyTest10 test does not pass.

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-
- 6) The BusyInsertionTest does not pass.
 - 7) Some of the RTL simulation log files are the incorrect log files
 - 8) The verification/bustest README has some references to obsolete tests

Implications

As description for issues 1) to 4).

Issue 5) The simulation log file produces the following error:

** Error: MPMCTR49 : HREADY0 is low for more than the specified number of clocks This error did not show up originally because the r2p2-00ltd0 version of this test was incorrectly running the "LatencyTest1.c"

Issue 6) The simulation log file produces the following error:

Error : Time 10957240 ns : MPMCTM19: Maximum time limit for the read access has exceeded Instance bench.u3MpmcTrickMem.uMpmcTrMemRdWrCtl.p_RdWrComb

This error did not show up originally because the r2p2-00ltd0 version of this test was incorrectly running the "LatencyTest9.c" for modelsim and "LatencyTest1.c" for LDV.

Issue 8)The "SimMakefile_<SIMULATOR>.mk" should be "DirSimMakefile_<SIMULATOR>.mk"

-> Tests - Arb2DyStWrRd and Arb3DyStWrRd - were not on the test-list with appropriate tbench number

-> References are made to the following redundant tests:

- > WriteMisMatchTest
- > StBufFlushTest
- > DyBufFlushTest
- > MemModelFlashTest
- > MultiPortRandomTest2

Affected Items:

LatencyTest10.c Mpmc.c MpmcTrAhbif.vhd BusyInsertionTest.c

Workaround

Workarounds are possible for the following items detailed in the Erratum Description:

(3) "verification/bustest/MultiPortDiffComb8.c" - Declare signals "LoBits" and "Bound" as follows: "int32 LoBits,Bound;"

(4) "verification/bustest/Mpmc.c" - Change the single occurrence of "Latencytest10" to "LatencyTest10"

(5) "verification/vhdl/MpmcTrAhbif.vhd" - Assign number of zeros for HRDATA o/p for "MPMCTrStCS" register to [31:7]. Previously [31:6].

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ERRATA - DOCUMENTATION

[324261](#): Incorrectly logged Errata 3.7

Status

Affects: product PL172_MPMC.

Fault raised on: revision r2p1 00LTD0 of PL172_MPMC

Fault status: Present in: r2p1-00LTD0, r2p2-00LTD0, r2p2-00LTD1, Fixed in r2p3-00RELO

Description

Issue 3.7 "A write following a burst transfer is ignored when no wait states are used" was incorrectly specified in the Errata document.

This defect does not apply to the product.

Implications

none

Workaround

none

[316115](#): Update to Errata 3.9

Status

Affects: product PL172_MPMC.

Fault raised on: revision r2p1 00LTD0 of PL172_MPMC

Fault status: Present in: r2p1-00LTD0, Still open.

Description

When reads are initiated to SDRAM/SyncFlash, in some conditions Data is Read from memory and not from AHB write buffer.

Summary:

This error occurs in the following conditions :

1. When all the buffers in the MPMC are full and you perform a single or incrementing write which is not of total size word. For example 3 byte INCR or 1 half word. That write is stored in the AHB write buffer waiting for either another write to fill it and make it a word in size, or a buffer to become empty for it to write to.

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OR

2. If an aligned WORD size Burst access(INCR4 or INCR8 or INCR16) is initiated to an address, and if the locations covered by the address are only partially present in the buffer (no problem if fully present or not present at all), and the buffer contents have not been posted to the memory yet.

If any one of the above condition occurs and a read is performed from that address, the MPMC will read from memory and not from the AHB write buffer. The data is not lost and will be written to memory later.

Symptoms: The data returned from a read will not be the data in the buffer but directly from memory. The data in the buffer will eventually be written to memory.

Implications

This is an update to Errata 3.9, published in the previous Errata document.

Workaround

All Reads must be either:

1. Access in SINGLE or INCR.
2. Not a Size of WORD, if the access is INCR4 or INCR8 or INCR16, and the burst address is an aligned address. Wrap Transfers will not have a problem.

Metal Fix:

A simple metal fix would be to tie the signal MemFetchQ to low in the file MpmcAhbif.v.

314789: Synthesis should be HCLK:MPMCCLK = 1:1

Status

Affects: product PL172_MPMC.

Fault raised on: revision r2p2 00LTD0 of PL172_MPMC

Fault status: Present in: r2p2-00LTD0,r2p2-00LTD1, Fixed in r2p3-00REL0

Description

The PL172 has been designed to be synthesised with a clock ratio of only 1:1. It does not support a ratio of 1:2 HCLK:MPMCCLK for synthesis. The documentation does not clearly state this.

Implications

none

Workaround

none

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309459: UserGuide States Use Denali*.mk Instead of DirSimMakefile*.mk

Status

Affects: product PL172_MPMC.
Fault raised on: revision r2p2 00LTD0 of PL172_MPMC
Fault status: Present in: r2p2-00LTD0,r2p2-00LTD1, Fixed in r2p3-00RELO

Description

UserGuide States Use Denali*.mk Instead of DirSimMakefile*.mk

Section 4.5.3 of the User Guide states that Denali.mk and Denali_LDV.mk files should be used. In fact these are obsolete.

The UserGuide should say use DirSimMakefile.mk and DirSimMakefile_LDV.mk

Implications

The following error will be received when denali*.mk files are used instead of the DirSimMakefiles:

```
make: Fatal error: Don't know how to make target `all'
```

```
Current working directory /PL172-BU-00000-r2p2 00ltd1/PL172_VC/mpmc_pl172/verification/verilog/reader
```

```
*** Error code 1
```

```
make: Fatal error: Command failed for target `all'
```

Workaround

Ignore the denali*.mk files present in the verification/\$HDL_SOURCE and instead use DirSimMakefile.mk and DirSimMakefile_LDV.mk when running simulations.